

4.4 Newton's laws of motion

Mechanics · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Resultant force equals mass times acceleration for a constant-mass body. In connected-particle models, write an equation for each body and use a common acceleration; tension is an internal interaction and cancels when treating both as one system. Resistive forces oppose motion, not necessarily velocity axes.

Worked example: follow the reasoning

For two masses 3 and 2 kg joined over a smooth pulley, $g=10$, $\text{acceleration}=(3-2)10/(3+2)=2 \text{ m s}^{-2}$.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A 4 kg mass hangs against a 1 kg mass over a smooth pulley, $g=10$. Find acceleration.

[3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

Driving force difference $(4-1)g=30\text{ N}$ [1]; total mass 5 kg [1]; $a=6\text{ m s}^{-2}$ toward 4 kg side [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.